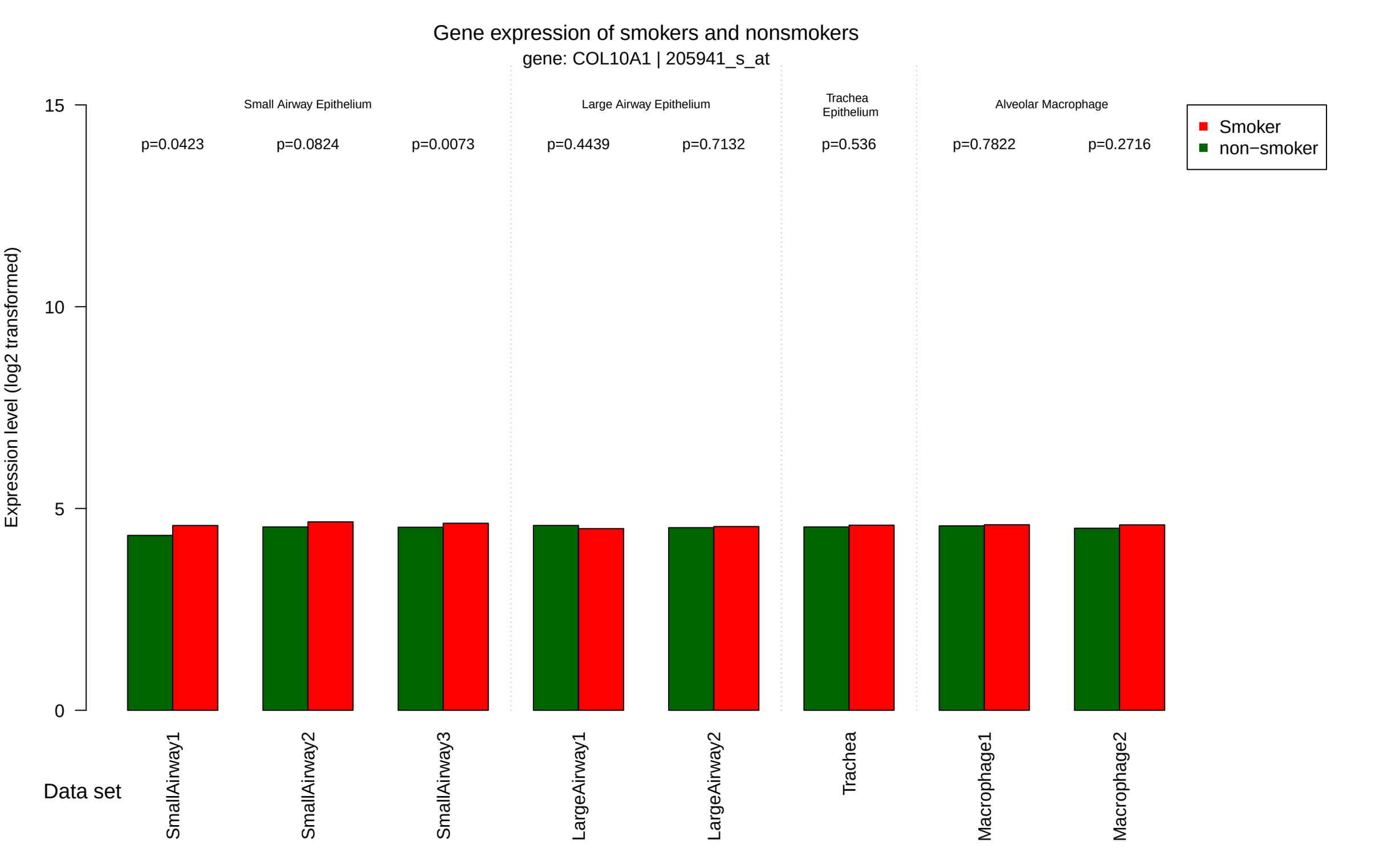
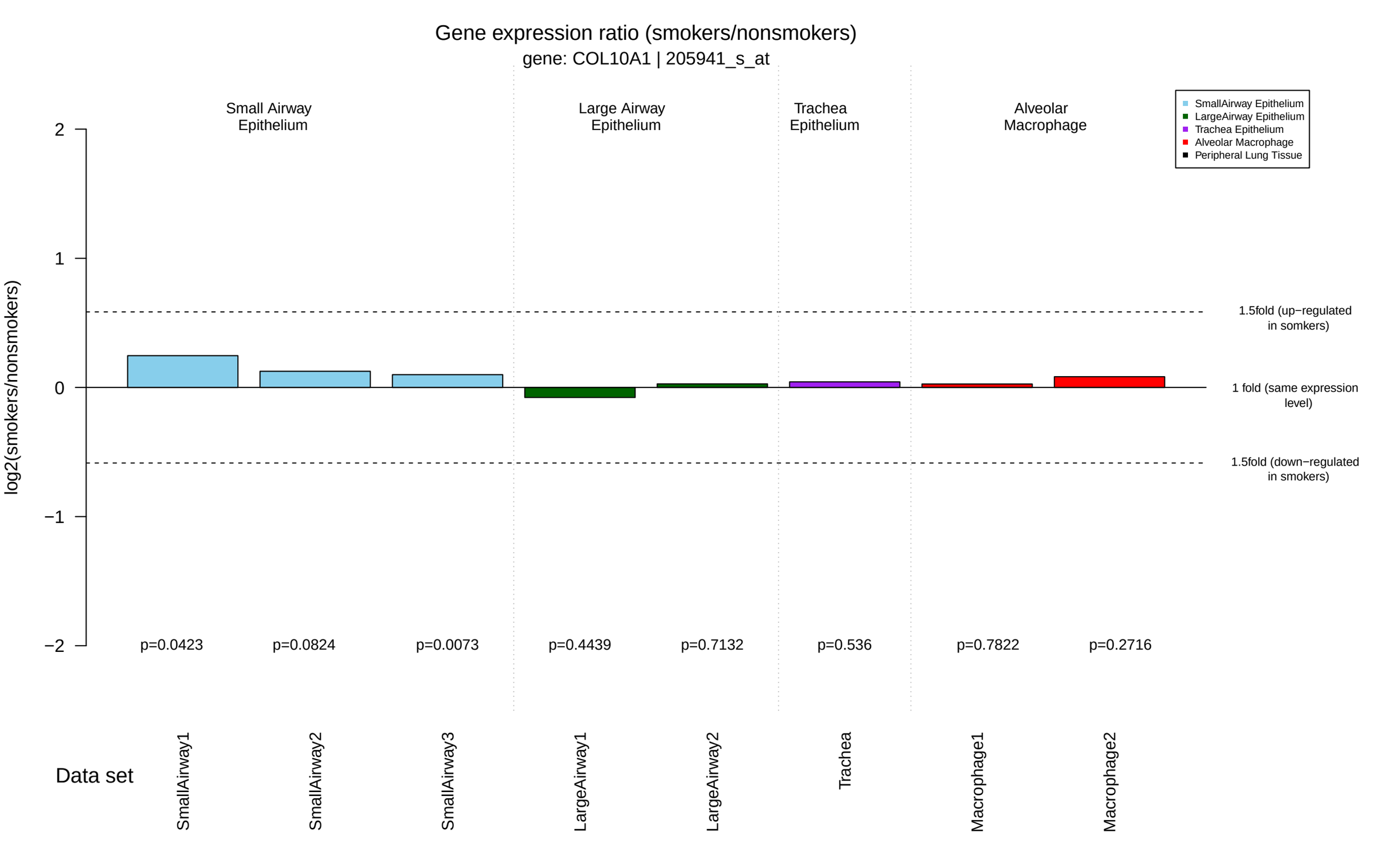
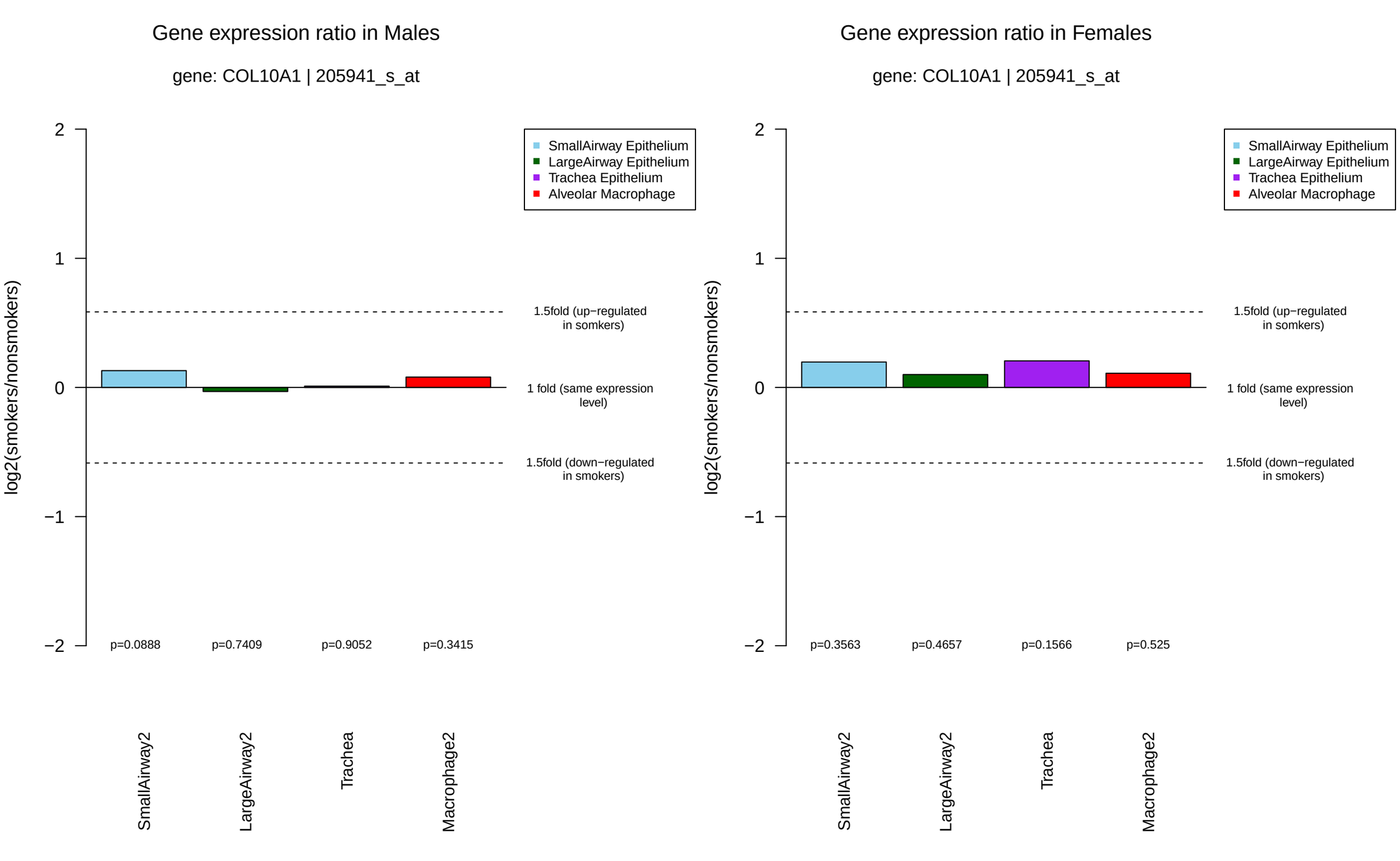
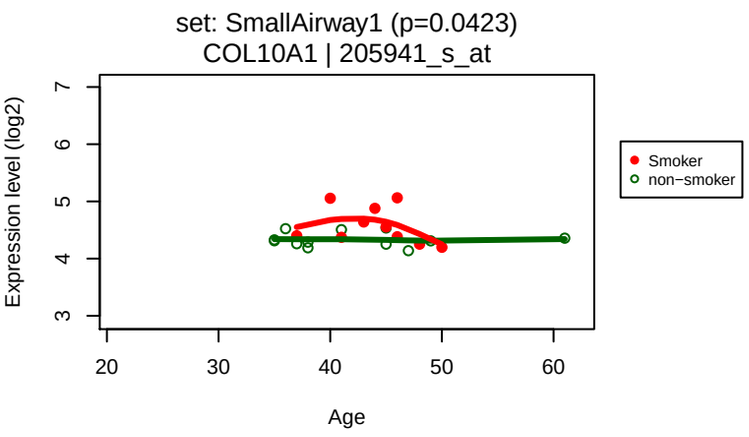
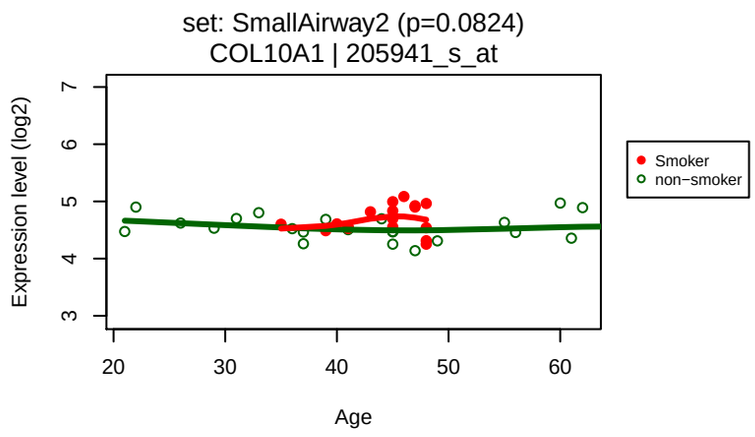
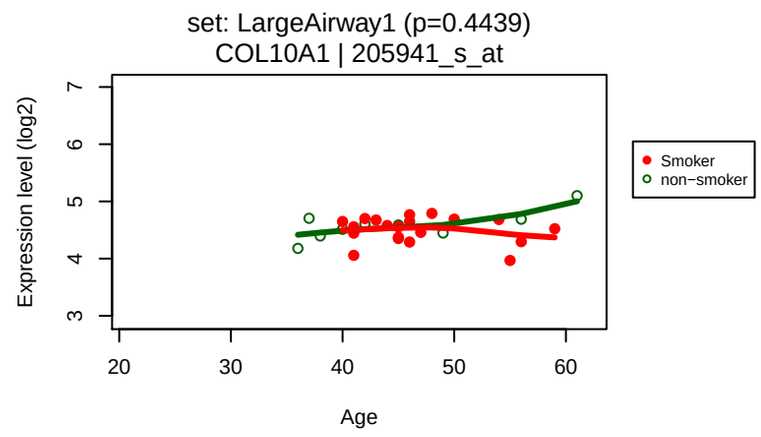

Gene expression of smokers and nonsmokers
gene: COL10A1 | 205941_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0423
p=0.0824
p=0.0073
p=0.4439
p=0.7132
p=0.536
p=0.7822
p=0.2716
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

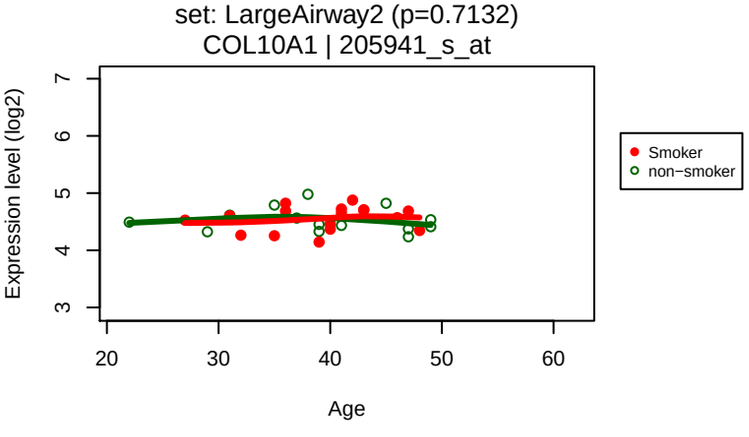

Gene expression ratio (smokers/nonsmokers)
gene: COL10A1 | 205941_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0423
p=0.0824
p=0.0073
p=0.4439
p=0.7132
p=0.536
p=0.7822
p=0.2716
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

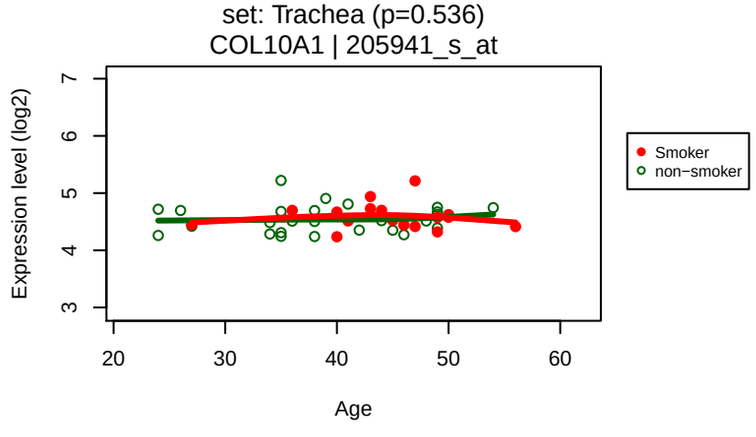

Gene expression ratio in Males
gene: COL10A1 | 205941_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0888
p=0.7409
p=0.9052
p=0.3415
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL10A1 | 205941_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3563
p=0.4657
p=0.1566
p=0.525
SmallAirway2
LargeAirway2
Trachea
Macrophage2

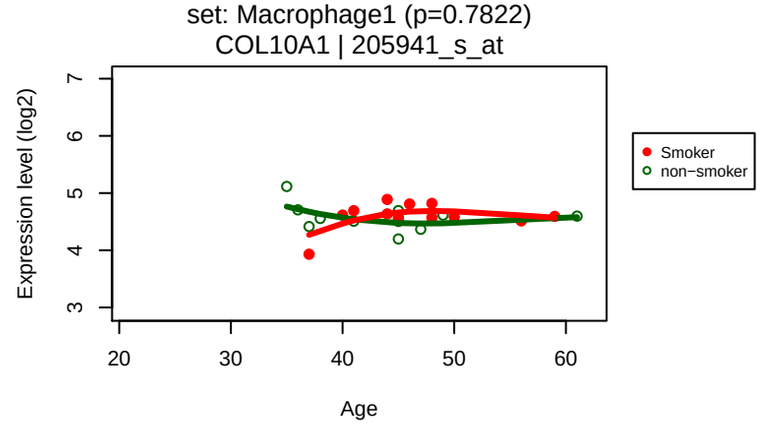

set: SmallAirway1 (p=0.0423)
COL10A1 | 205941_s_at
Expression level (log2)
Age
Smoker
non−smoker

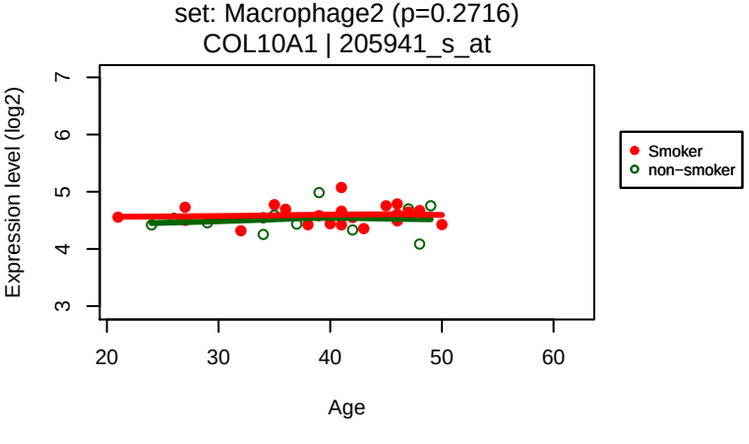

set: SmallAirway2 (p=0.0824)
COL10A1 | 205941_s_at
Expression level (log2)
Age
Smoker
non−smoker

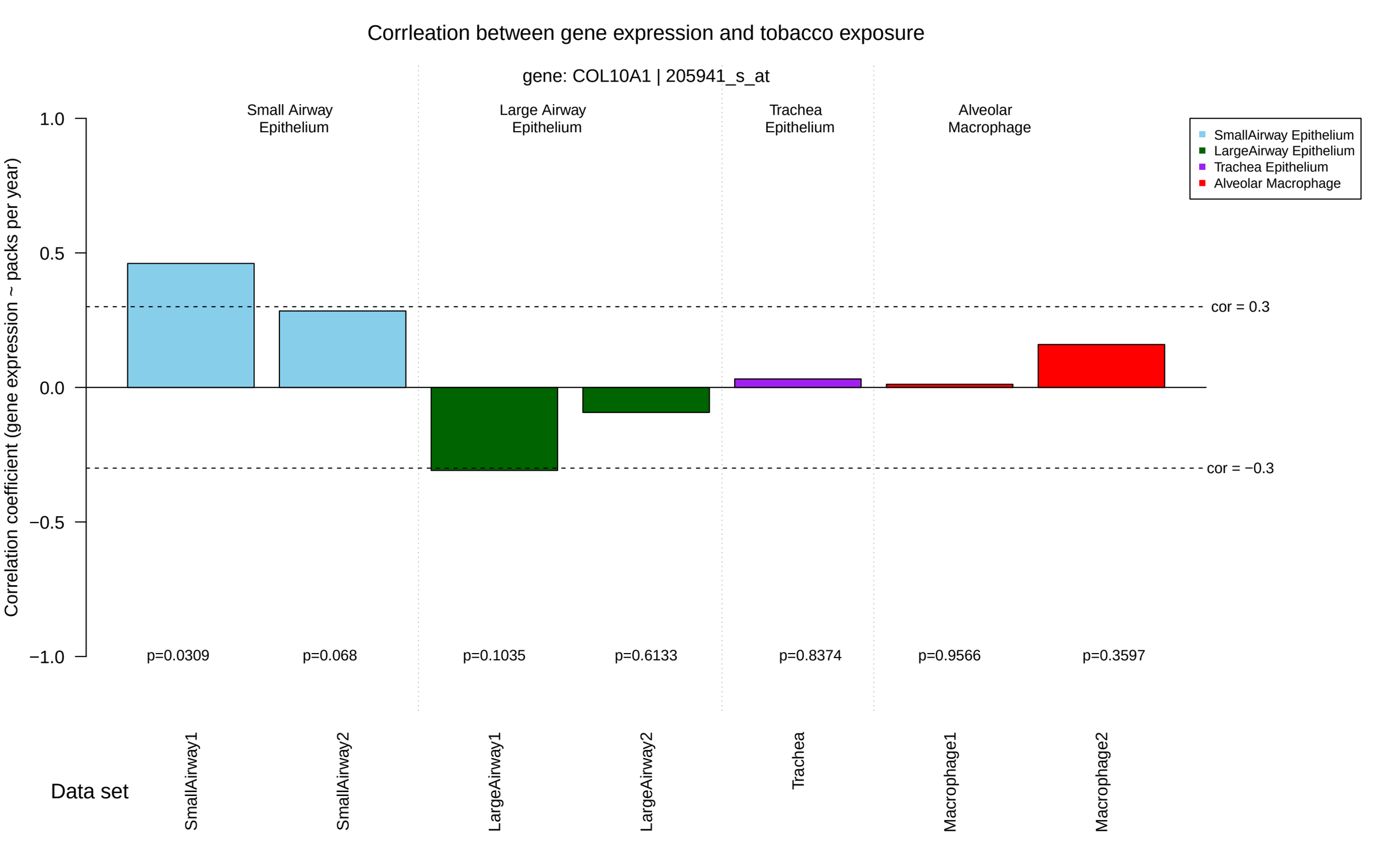

set: LargeAirway1 (p=0.4439)
COL10A1 | 205941_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7132)
COL10A1 | 205941_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.536)
COL10A1 | 205941_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7822)
COL10A1 | 205941_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2716)
COL10A1 | 205941_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL10A1 | 205941_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0309
p=0.068
p=0.1035
p=0.6133
p=0.8374
p=0.9566
p=0.3597
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2